

Track 1 — Vehicle Selection for Data Acquisition

Which India-market cars to use for OBD-II data collection and validation

Car Automation IoT Internship

1. Purpose & The Honest Starting Point

This document identifies which vehicles are suitable for our OBD-II data acquisition. It is a starting point for the bench test, not a final answer — because the single most important fact about vehicle OBD-II support is this:

Real, per-model confirmation of which parameters a car returns does not exist in public sources. It can only be confirmed by plugging in and logging the actual responses. No online claim, datasheet, or AI answer counts as confirmation.

So this list ranks vehicles by how likely they are to work and how easy they are to borrow — and the chosen car is then confirmed by a bench scan.

2. The Key Point: Standard Data Works Almost Everywhere

For the standard parameters — RPM, speed, coolant, engine load, throttle, intake temperature, fuel level, and diagnostic trouble codes — almost any petrol car in India from 2010 onward works. These are mandated, standardised OBD-II parameters.

The differences between cars are only at the margins:

- MAF (air-flow) — needed to calculate fuel mileage — is present on most Maruti/Hyundai engines but often absent on Toyota engines (which use a MAP sensor instead). This is the one field where car choice genuinely matters.
- The manufacturer-specific fields (AC status, brake, clutch %) may be unavailable on any car — that is an OBD-standard limitation, not a car-selection problem. No car choice fixes it; only the bench scan tells us.

Therefore the deciding factor is availability. A car we can plug into ten times while debugging is worth far more than a “cleaner on paper” car we can borrow once. Pick whichever compliant car the team can most easily and repeatedly access.

3. Tier 1 — Safest Picks (Recommended)

Standard parameters reliable, common in India, easy to borrow. Start here.

Vehicle	Year / fuel	Why it's a good pick
Maruti Swift / Dzire / Baleno / WagonR	2018+ BS-VI petrol	Best overall pick. BS-VI Marutis are the generation explicitly regarded as standards-compliant CAN. Standard PIDs reliable, MAF generally present. Most borrowable car in India — very likely someone on the team or in the office has one.
Hyundai i20 / Verna	2015+ petrol	CAN-based, standard PIDs read well per owner reports, MAF generally present. Very common.
Hyundai Creta / Venue	2019+ BS-VI petrol	Modern Hyundai/Kia CAN platform, strong standard compliance expected, extremely common SUV — high availability.
Tata Nexon / Tiago / Altroz	2020+ BS-VI petrol	Petrol variants use the Bosch ME17.9 ECU, standard-tool-friendly. Buy/borrow the PETROL variant — diesel uses a newer ECU (CAN-FD) needing a fancier scanner.
Honda City / Amaze / Jazz	2014+ petrol	Standard PIDs work; Honda dealership access is common if verification is wanted. Some fuel-trim PIDs can be quirky, but the core set is fine.

4. Tier 2 — Works, With Caveats

Vehicle	Caveat
Toyota Innova Crysta / Glanza / Etios	Standard PIDs (RPM, speed, coolant) fine, but Toyota engines often OMIT MAF (0x10), using a MAP sensor instead — so fuel-mileage (which needs MAF) may not compute. Everything else works.
Older Maruti (Swift/Celerio 2015–2017 pre-facelift)	Some early units respond “service not supported”, or use a non-CAN native protocol (ISO 9141 K-line) under the hood. Riskier — post-2018 Maruti is the safer bet.
Mahindra XUV / Scorpio; older diesels	Standard emissions PIDs still read, but expect more manufacturer-specific fields to be null. Diesels can add quirks.

5. Tier 3 — Avoid for the First Test

Avoid	Why
Pre-2010 cars	May predate the OBD-II CAN mandate in India; some use older/non-standard protocols a generic reader cannot decode cleanly.
CAN-FD diesels (some 2021+ Tata / newer premium)	Need a CAN-FD-capable adapter, not a basic ELM327.
Very new premium / luxury / EV	Often gate the OBD port behind security and are less openly documented.

6. What Data to Expect From Any of These

Regardless of which Tier 1 car is chosen, the fields split like this:

Field(s)	Expectation	Note
RPM, speed, coolant, engine load, throttle, intake temp, fuel level, DTC count	Expected to work	Standard OBD-II parameters — reliable on any Tier 1 car.
MAF (air flow)	Car-dependent	Present on most Maruti/Hyundai, often absent on Toyota. Fuel mileage depends on it.
AC status, brake, clutch %	Likely null	Manufacturer-specific — may be unavailable on ANY car. Not fixable by car choice.
Gear, fuel mileage	Calculated	Derived in our own code from other values, not read from the car.

7. Recommendation

Primary: a 2018+ BS-VI petrol Maruti (Swift / Dzire / Baleno / WagonR) — the best combination of clean standard compliance and, crucially, real-world availability to borrow repeatedly.

Backup: a Hyundai i20 or Creta (2019+ petrol) — equally strong compliance, also very common.

Deciding rule: since standard data works on all Tier 1 cars, pick whichever one the team can physically access most easily and often. Availability beats paper-compliance.

8. Next Step — Confirm By Bench Scan

Once a car is available, confirm it before committing:

- Connect the ELM327 and run the supported-PID probe (0100) to list what the car exposes.
- Request each target PID and log the actual response — a valid data response (good) or a “not supported” response.
- Record, for that specific car, which fields are confirmed and which are null.
- That logged output — not any claim in this document — becomes the real compliance record for sign-off.